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PRAHARI

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Module 01 — PRAHARI

Lectures Stakeholder briefing

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Outcomes

- Tell the school-notebook story: cameras already exist, the watchlist is unused
- Name the hybrid (registry + viewing + event bus) and say why central recording is later
- Trace one frame through capture, analyse(), matcher, alert, CSV
- Separate MEASURED laptop facts from DESIGN TARGET 80k / 5–6 Cr numbers
- State the FRS law: enrolled gallery on Own cameras, never Paldi Circle



Who this briefing is for

- Operators, architects, and stakeholders
- Voice: a lecturer naming parts a ten-year-old can follow
- Every number is MEASURED, DESIGN TARGET, or unused (CONJECTURED)
- Product of Yushu Excellence Technologies Pvt. Ltd.
- Collaborator: Amit Dua



The school-notebook analogy

- 26 schools. Each bought a different cupboard for tapes
- Some cupboards keep 7 days. Some keep 15
- Head office has a notebook of stolen bicycles and missing children
- Nobody in the camera room opens that notebook
- That is Gujarat CCTV: 26 departments, analog and IP, 1,000 km
- Watchlists exist (VAHAN, SARTHI, eGujCop, AFIS, NAFIS) and sit unused



What the State already has

- Independent CCTV estates. No statewide camera census
- Cloud NVR in some estates, local storage in others
- Sites: border districts, Valsad, Dahod, Somnath, Jamnagar, Dwarka
- The design goal: use existing infrastructure first
- A new cupboard for every tape is the wrong first move



Four operating facts

- Heterogeneous infrastructure: vendors, VMS, AMC, formats, protocols
- Geographical dispersion of about 1,000 km
- Unified analytics and event handling across onboarded cameras
- New cameras and departments without a redesign

Core goal: use existing infrastructure to the maximum practical extent.



Architecture options

Layer	Name	School picture
1	Registry and GIS	Roll call. No live picture required
2	Unified viewing	One window, many classrooms
3	VMS federation	Post office between cupboards
4	Central VMS	One giant cupboard for the city
H	Hybrid	Combine two or more

- Model 1 is mandatory foundation for 2, 3, and 4



Options we rejected

- Model 1 only: a roll call does not catch a stolen car
- Viewing only: watching without a watchlist does not catch a stolen car
- Full Model 3 SDK zoo: seven days cannot wrap every vendor
- Model 4 statewide: we cannot host 80,000 recordings (DESIGN TARGET)
- Wild FRS on Paldi Circle: unknown members of the public
- A ministry biometric or vehicle API: we have none. Claiming one is false



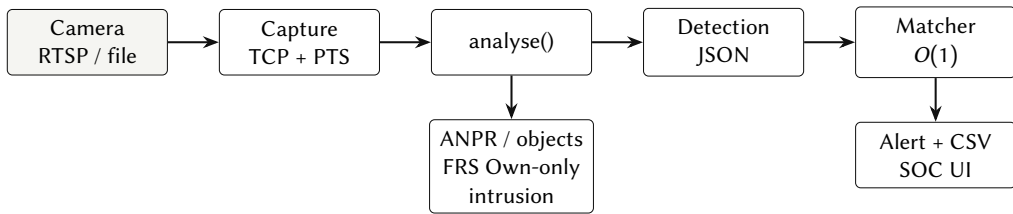
Chosen model

Layer	Architecture	This tree
Census, GIS, health, gaps	1	now
Unified viewing + analytics	2	now
Event bus, watchlist, track	thin 3	now
Central VMS recording	4	Phase-2, selected cameras

- A new vendor is one adapter on RTSP / HLS / WHEP / ONVIF
- Detection JSON does not change if the bus later becomes Kafka



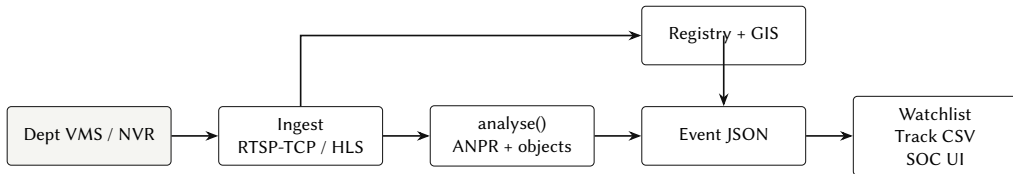
How a picture becomes a red card



- Capture stamps PTS. analyse() samples about 1 fps
- Matcher is $O(1)$. Stolen plate raises CRITICAL. CSV keeps timestamps



Logical architecture



- PoC bus is an in-process WebSocket. The JSON is the contract
- Analytics samples 1 frame per second, not a 25 fps GOP



What operators demonstrate

- Onboard about 50 heterogeneous cameras from the Resources page
- Integrate live feeds. Enable monitoring and video analytics
- Identify a designated vehicle registration number across the grid
- Reconstruct route, timestamps, locations. Plot on GIS
- Match detections to a representative watchlist. Raise automated alerts
- Output: working platform, not a mock. CSV / report with timestamps



Live Sentinel contract (03 Sep 2026)

- Portal: `cctv.corp8.cloud` after access-password login
- Catalogue: GET `/cameras.json` returns 30 rows {id, name}
- `/api/ingest` is HTTP 404 on this host
- HLS: `https://cctv.corp8.cloud/<id>/index.m3u8` (cookie + browser UA)
- RTSP: `rtsp://103.250.160.189:8554/stream/<id>` (TCP)
- One second of video takes one second to arrive. There is no file download



Integrator laws in the capture client

- Set `OPENCV_FFMPEG_CAPTURE_OPTIONS=rtsp_transport;tcp` before `import cv2`
- Event time is PTS (`CAP_PROP_POS_MSEC`), never arrival time, never declared FPS
- Reconnect backoff starts at 2 s and caps at 30 s
- Decoder warnings at join are logged, not fatal
- PTS rewind or gap > 5 s is a scene cut (feeds loop)
- Consume only. Do not `wget /stream/<id>`
- Tests: `tests/test_integrator_laws.py`



Why those laws (ten-year-old version)

- UDP across a firewall arrives as a broken picture that looks like a bad model
- After join, extra frames may arrive fast. Wall-clock time invents flying cars
- Feeds restart. Reconnecting in a tight loop is shouting at a locked door
- The tape loops. At the splice, forget the old object ids
- A downloaded /stream file is a fake dataset



MEASURED: government feed (03 Sep 2026)

- Synced 30 catalogue cameras into the registry (cam01 ...cam30)
- RTSP-TCP probe live: cam04 Paldi Circle, cam01 Chimanbhai Bridge
- First live frame: 03_Data/recordings/first_live_frame.png
- Size 1920×1080, PTS 1080 ms. UDP SETUP 461 then TCP succeeds
- Manifest has no live flag and no lat/lon. Pins stay off the map
- Open tiles from the Cameras table, not from a 0,0 marker



MEASURED: designated vehicle

Time	Camera	Location
06:12	CAM-VAL-001	Valsad NH-48 Toll
07:48	CAM-SUR-001	Surat Adajan Circle
09:05	CAM-AHD-004	Narol-Naroda Hwy
09:22	CAM-AHD-001	SG Highway Junction
09:51	CAM-GNR-003	Koba Circle
10:04	CAM-GNR-001	CH-0 Sector 21 GNR
11:40 IST	cam04	Paldi Circle (live confirm)

- Plate GJ01AB1234, watchlist WL-001 / STOLEN
- Live confirm appends. It does not drop the six seed points

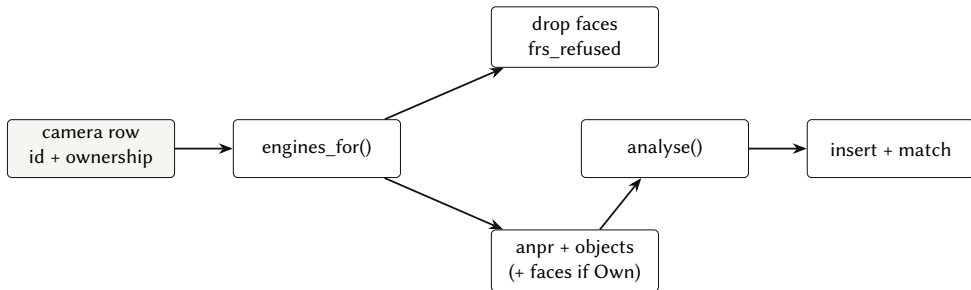


`analyse(): one door`

- Default engines: anpr, objects
- Own cameras may add faces
- cam plus digits, and any Gov camera, never run FRS
- Godown CAM-FCS-001 also runs person-in-ROI intrusion
- Plate fields stay frozen. New fields are additive
- Ingest writes crops, inserts SQLite, then calls the matcher



engines_for(): who may run faces



- Log line frs_refused is the control, not a post-filter
- FaceAnalyzer is not constructed on Paldi Circle
- POST /api/ingest/frame stays ANPR-only



ANPR: options

- Train a detector from scratch: no labelled Gujarat set in seven days
- Require GPU YOLO now: GPU count MEASURED 0 on this laptop
- Morphology + Tesseract behind recognize(): CPU, swappable
- Operator confirm: a readable plate cannot fail a live demo
- Confirm writes `source=operator_confirm`. Never labelled ANPR



ANPR: mechanism

- Sampled frame → plate box → Tesseract → Indian normaliser
- Regex after stripping space and hyphen: `^[A-Z]{2}[0-9]{1,2}[A-Z]{1,3}[0-9]{3,4}$`
- Interface: `recognize(frame) -> {plate, plate_raw, confidence, crop}`
- Missing Tesseract binary: empty plate, no crash
- MEASURED 03 Sep: confirm on cam04, confidence 1.0
- Optional: `ANPR_ENGINE=yolo` vehicle-crop then Tesseract, then fallback



Objects: options

- YOLOv8n as the only path: extra dependency, tests would skip
- OpenCV DNN with large weights: allowed later, not required for pytest
- Skin-tone blob fallback: a rectangle $\geq 8\%$ of the frame is a person
- Skipping objects: eval area 05 names person and vehicle detection
- Classes kept: person, car, motorcycle, bus, truck, bicycle



Objects: mechanism

- Default `OBJECT_ENGINE=opencv`: blob, then optional DNN
- Optional `OBJECT_ENGINE=yolo`: COCO subset including person (class 0)
- Missing ultralytics or weights: blob fallback, no crash
- Each box gets per-camera `track_id` (IoU default)
- Scene cut calls `objects.reset`
- Evaluator artefact: `GET /api/objects/report.csv`



Intrusion on a godown

- Food and Civil Supplies watches a warehouse, not a highway
- ROI lives in `cameras.extra_json`
- Person box overlapping the ROI by $\geq 30\%$ of its area fires
- Event: `entity_type=intrusion, CRITICAL`
- Same camera within 120 s increments a counter
- No fourth neural net. Intrusion wraps object boxes



FRS: options we refused

- No FRS: HLD text and eval 05 name recognition
- Wild FRS on Paldi Circle: unknown members of the public
- Live AFIS / NAFIS: we have no ministry API. Claiming one is false
- Enrolled gallery on Own cameras: consented adults or synthetic fixtures
- Human confirm-face exists, same safety net as plate confirm



FRS law (non-negotiable)

- Own-feed may show a gallery match on CAM-OWN-001
- Government CCTV of unknown people never runs FRS
- Tests post a face-like still to cam04 and expect zero person events
- Gallery JSON has no embeddings
- Spoken line: representative person watchlist, not a NAFIS join



FRS: default histogram mechanism

- Crops live under `data/faces/{gallery_id}/`
- Match: colour histogram of a 64×64 crop, correlation score
- Threshold: `FACE_MATCH_MIN_CONFIDENCE` (default 0.55)
- WL-004 fires on `face_id`, not on an empty plate
- Synthetic fixtures are ellipses with unique noise
- Never a minor. Never a crop from `cctv.corp8.cloud`



Optional vision engines

- FaceNet when `FACE_ENGINE=facenet`, Own cameras only
- YOLO when `OBJECT_ENGINE` or `ANPR_ENGINE=yolo` (person class included)
- ByteTrack when `TRACK_ENGINE=bytetrack`; scene cut still resets
- Next-camera predict: frequency + distance, not Kalman
- Never Paldi FRS. No CodeFormer in this PoC
- Default remains histogram + blob + Tesseract + IoU
- This repository is the product tree: [amitduabits/PRAHARI](#)



FaceNet door (optional)

- Lazy FaceAnalyzer: MTCNN boxes + 512-d InceptionResnetV1
- Cosine against .npy files next to gallery JPEGs
- Torch is imported inside `__init__`, never at package import
- Missing torch: histogram fallback, no crash (T-V01)
- Constructed only after `engines_for` allows faces (T-V04)
- `requirements-vision.txt` is optional. Pytest does not need it



YOLO and ByteTrack doors (optional)

- Object YOLO includes person. Vehicle YOLO for ANPR does not
- Weights: `app/models_data/yolov8n.pt`, gitignored
- Fetch script exists. Pytest does not download
- ByteTrack writes object `track_id`. IoU remains default
- GET `/api/track/{plate}` stays the designated-vehicle test
- GPU count MEASURED 0 on this laptop



Crop honesty

- Always write `crop_uri_original`
- Matcher sees original pixels only
- Optional enhanced copy: `enhancement_method=cubic_upscale` or `none`
- Face crop with `is_ai_reconstructed=1` → `pending_review`
- Reconstructed faces do not auto-fire CRITICAL
- Zero branded reconstructor names in app/ (T-V11)



Watchlist and alerts

- Representative sources. No live ministry API in the PoC
- $O(1)$ on plate and on face_id / gallery_id
- CRITICAL stolen/wanted/intrusion. HIGH blacklist or person associate
- Same entity + same camera within 120 s increments a counter
- WebSocket /ws/alerts plus 5 s poll fallback
- WL-004 fires without a plate. Person cards show a name, not a blank
- MEASURED: confirm on cam04 raised CRITICAL for GJ01AB1234



Person enroll and pending review

- Watchlist form: Enroll Missing/Wanted Person
- Posts `/api/faces/enroll`. Viewer still 403 (T-V10)
- Seven tabs. No eighth top-level tab
- Alerts renderer shows the word `pending_review`
- Ack still works
- Do not enroll from a Sentinel still



Vehicle track is plate GIS, not Re-ID

- GET /api/track/GJ01AB1234: six seed cameras in order
- Live confirm appends. Seed ids stay
- Teleport flag if implied speed > 180 km/h (information, not a fail)
- CSV is the operator artefact
- ByteTrack does not replace this API



Next-camera predict (bonus)

- GET /api/predict/{plate}, auth required
- Rank historical transitions out of the last camera
- If none: GIS neighbours by haversine
- Method field: frequency+distance
- Not Kalman. Not Re-ID
- Track tab button: Next cameras



Keyword query (optional)

- POST /api/query extracts plate, camera id, object class
- Response includes engine: keyword_rules
- Regular expressions, not a language model
- T-V09 forbids `nlp` and `llm` in the body
- Do not put “natural language AI” on a slide



Registry, GIS, gaps

- CSV, form, REST, catalogue sync
- Dahod GSRTC is seeded offline on purpose
- Short retention: Food and Civil Supplies 7-day cameras
- Private-permitted mall requires `consent=true`
- `home.viewer` never sees that mall
- Live catalogue has no coordinates. Open `cam04` from the table



Seven tabs

- Operations: map, health pins, wall of at most four tiles
- Cameras: table Open tile (catalogue has no GIS)
- Vehicle Track: reconstruct, CSV, Next cameras
- Alerts: CRITICAL queue, pending_review, Ack
- Watchlist: representative rows, person enroll
- Onboard: ANPR this still, Analyse this still, confirm
- Analytics & Gaps: object CSV, short retention, Dahod

Capability map



Capability	PRAHARI now
Live ingest	Catalogue cameras. RTSP-TCP frame. Confirm + CSV on cam04
Architecture	HLD. Open adapters. Central recording not faked
Working platform	FastAPI on :8080. Own-feed file + live catalogue
Analytics	ANPR/confirm + objects CSV + godown intrusion + own-feed gallery FRS
Scale	1 fps regional DESIGN TARGET in HLD §5 / §10
Also on camera	Track, objects, intrusion, lawful FRS, RBAC, gaps, WS



Test families (04 Sep 2026)

- Default engines: 88 passed, 4 skipped. `audit_gate.py` PASS
- Integrator: TCP, no CAP_PROP_FPS, scene-cut resets track ids
- ANPR: normaliser, junk drop, confirm $\neq$ source=anpr
- Objects: person blob inserts. YOLO without weights does not crash
- FRS: same gallery matches. cam04 emits zero person events
- Matcher: stolen, blacklist, WL-004 without a plate, 120 s dedupe
- Security: viewer 403 on enroll. Auditor 403 on confirm-face. No RTSP in JSON



Vision tests T-V01 to T-V11

- T-V01 / T-V02: missing torch or ultralytics → histogram / blob, no crash
- T-V03 / T-V05 / T-V06: skip unless extras and weights exist
- T-V04: FaceAnalyzer not constructed on cam04
- T-V07: original crop uri. Reconstructed → pending_review
- T-V08: predict returns a list. T-V09: keyword_rules
- T-V10: Enroll in index. Viewer 403
- T-V11: zero forbidden model names in app/



Experiments: MEASURED versus SKIPPED

- E-O1 person blob, E-F1 gallery, E-I1 intrusion, E-W1 track ≥ 6 : MEASURED
- E-S1 fifth capture rejected. E-S6 GPU 0. MEASURED
- E-A1 Tesseract fixture: SKIPPED if binary absent (confirm covers demo)
- E-G1 live catalogue: SKIPPED if SENTINEL_HOST empty
- E-V2 / E-V3 FaceNet and YOLO: SKIPPED without extras
- E-V4 cam04 + FaceNet still 0 person events: MEASURED
- Log: 05_Output/experiments/EXPERIMENT_LOG.md



Eighty thousand cameras (DESIGN TARGET)

- About 45,000 public-domain cameras run analytics at 1 fps, 720p crop ≈ 80 KB
- Naive centre: $45,000 \times 80 \text{ KB/s} \approx 3.6 \text{ GB/s}$
- Five regions (Ahmedabad, Surat, Rajkot, Vadodara, Bhuj): $\approx 720 \text{ MB/s}$ each
- SOC wall pulls 16–64 HLS tiles. Everything else is metadata + crops
- These numbers are design targets, not measured laptop throughput



MEASURED on this laptop (04 Sep 2026)

- Four open file captures succeed. The fifth is rejected
- Mean JPEG from eight own-feed stills: 41 KB
- /api/health p99: 9.4 ms over 50 calls
- GPU count: 0. Regional GPUs stay DESIGN TARGET
- Statewide 80 KB crop remains DESIGN TARGET (ratio $1.9\times$)



Cost of the intelligence plane (DESIGN TARGET)

Item (statewide, per year)	INR
5 regional GPU nodes (3-yr amortised)	1.8–2.4 Cr
Object storage for crops + selected video	0.8–1.2 Cr
SWAN / leased bandwidth top-up	0.6–1.0 Cr
Integration + SOC ops (12 engineers)	1.5 Cr
Total intelligence plane	~5–6 Cr

- This is not the bill for replacing 26 VMS estates



Capture internals

- StreamSession is the only OpenCV capture holder
- TCP flag is set before `import cv2`
- Event time is PTS. Wall clock is only for reconnect backoff
- MAX_OPEN_CAPTURES default 4. Fifth open raises
- Scene cut: PTS rewind or gap > 5 s. Object and face tracks reset
- File protocol is `own_feed.mp4` on CAM-OWN-001



RBAC matrix

User	Role	Sees	Write
judge	soc_operator	all	yes
admin	superadmin	all	yes
home.viewer	dept_viewer	Home only	403
auditor	auditor	all	403

- Mall camera is hidden from Home viewer
- Enroll and confirm-face are write routes



Path jail and HMAC tiles

- Crops resolve inside media roots. . . is rejected
- Stream token: HMAC over camera, actor, expiry. TTL 60 s
- Full digest, compare_digest. Not a sliced hash
- Browser never receives rtsp://
- hls.min.js is vendored. No jsDelivr
- Sentinel cookie stays on the server



What we took, what we refused

- Take: FaceNet, YOLO with person, ByteTrack, enroll form, original crops, predict
- Refuse: FaceAnalyzer on every camera
- Refuse: 80k footer without DESIGN TARGET
- Refuse: catalogue /api/ingest (live 404)
- Refuse: torch in default requirements
- Refuse: branded reconstructor names, Wikipedia tests, changing origin



Security in the PoC

- Roles: superadmin, soc_operator, dept_viewer, auditor
- home.viewer sees Home cameras only; private-permitted mall is hidden
- HLS/file tiles use a 60 s HMAC token. Raw RTSP never reaches the browser
- Sentinel cookie stays on the server. The browser never sees it
- Private-permitted onboard requires consent=true
- Audit: onboard, watchlist, ack, report, plate confirm, face confirm



Stack (open source, as required)

Piece	PoC	Later swap
API	FastAPI + Uvicorn	same contract
UI	Leaflet + vanilla JS	React if needed
Store	SQLite	PostgreSQL + PostGIS
Capture	OpenCV + FFmpeg, RTSP TCP	DeepStream
ANPR	morphology + Tesseract	YOLO + PaddleOCR
Alerts	in-process WebSocket	Redis / Kafka



What is done, what is still human

- Done: platform, HLD, this deck, live catalogue sync, RTSP frame, confirm CSV
- Done: objects, intrusion, lawful FRS with gov refuse, optional engine pack
- Done: GitHub github.com/amitduabits/PRAHARI
- Human: own-feed YouTube ≤ 3 min. Gov-feed YouTube ≤ 3 min, no FRS
- Human: Drive Anyone+Viewer for gov_feed_plates.csv
- Human: Tesseract on PATH if OCR should appear on camera
- Stop product edits after the portal receipt on 07 Sep



Demo run (3 minutes)

- Open <http://127.0.0.1:8080>. Login judge
- Operations map: seeded pins, Dahod offline on purpose
- Cameras table: Open tile on cam04 or cam01
- Vehicle Track: reconstruct GJ01AB1234, download CSV
- Alerts: CRITICAL queue, Ack writes audit
- Onboard: Analyse this still, or Operator confirm if OCR misses
- Gaps: short retention on Food and Civil Supplies